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Enhancing NLP for Low-Resource Languages using Cross-Lingual Transfer and Few-Shot Learning

Devalla Bhaskar Ganesh
Computer Science and Engineering
Koneru Lakshmaiah Education Foundation
Vaddeswaram,India
devallabhaskarganesh@gmail.com

Greeshma Yaluru
Software Engineer
Populus Group
Pittsburgh,United States of America
gyaluru@gmail.com

Dokuparthi Nilesh
Computer Science and Engineering
Koneru Lakshmaiah Education Foundation
Vaddeswaram,India
dnileshmohan@gmail.com

Adithya Nagendra Gummadi
Computer Science and Engineering
Koneru Lakshmaiah Education Foundation
Vaddeswaram,India
2100031347cseh@gmail.com

Chittela Venkata Sai Tarun Reddy
Computer Science and Engineering
Koneru Lakshmaiah Education Foundation
Vaddeswaram,India
tarunreddych0@gmail.com

Suryakanth V Gangashetty
Computer Science and Engineering
Koneru Lakshmaiah Education Foundation
Vaddeswaram,India
svg@kluniversity.in

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***Abstract*—Much progress in NLP has happened in recent years and most of that progress is driven by data-rich languages. The absence of sufficient data in many low-resource languages causes many issues in the development of NLP tools. The framework introduced in this study helps overcome this limit, as it merges methods for language learning and minimal data training for supporting NLP in low-resource languages. Relying on the XLM-R multilingual language model, we help information transfer from big languages to similar smaller counterparts. When we arrange our tasks with premise-hypothesis design, we can more easily use language understanding systems and relate the different tasks we learn. To help further, few-shot learning is applied which allows the model to work with very little target language data. Our results indicate that learned models outperform others when there are few examples, since they work well in unseen circumstances. As a result, this project closes the digital divide among languages by creating a way to help NLP support underrepresented languages more effectively and economically. The framework supports and shapes the study of AI that is inclusive and can handle several languages.**

Keywords—Natural Language Processing, Few-Shot Learning, Low-Resource Languages, XLM-R, Cross-Lingual Transfer Learning

I. Introduction

There have been major advances in Natural Language Processing (NLP) in recent years mainly thanks to having a lot of data and progress in deep learning. Advances in AI have mainly helped languages with lots of data such as English and have created highly precise systems for work such as classifying texts, translating and answering questions. Nonetheless, many languages worldwide are normally absent from computational research because not enough of their data has been digitized and annotated. Such a difference points out the urgent need to help NLP technologies benefit low-resource languages [1].

Cross-lingual transfer learning focuses on how the problem can be solved by using one or more high-resource languages using transferring techniques on low-resource languages. In this sense, XLM-RoBERTa and other multilingual pretrained models offer great potential [2]. Their training on diverse corpora enables them to develop generalized language representations beneficial for multiple languages. Such capability can be exploited to minimize the task-specific annotated data by tailoring models to be constructed using data from other languages to be implemented for tasks in a target language.

In addition to cross-lingual methods, few-shot learning has become an effective way to train models with limited quantities of data. Few-shot learning mimics human learning by allowing models to learn and execute tasks based on a small number of examples. For the scenario of low-resource languages, the method is priceless, as it reduces the necessity of collecting and annotating a large amount of data, which requires time and money [3]. With the application of few-shot learning and cross-lingual transfer, it is now possible to create NLP models which are data-efficient as well as low-resource competent.

This study aims to apply a hybrid approach of cross-lingual transfer and few-shot learning to enhance the performance of NLP tasks in low-resource languages. In particular, we investigate sequence classification and natural language inference tasks through the XLM-R model. The methodology involves preparing multilingual datasets, creating premise-hypothesis input formats, and fine-tuning the model using small data from the target language [4]. We also integrate a friendly web interface to allow the system to be easily accessible and interactive for real-time use.

Our results show that this hybrid approach can attain competitive performance even with limited resources. The effectiveness of this framework proves its value for facilitating linguistic diversity in NLP tasks and addressing the digital divide. This research not only helps advance resource-constrained NLP models but also encourages the establishment of inclusive technology for underrepresented languages globally [5].

II. Literature Review

A. Overview of Low-Resource Language Challenges

Low-resource languages, by definition, don't possess the linguistic resources—annotated corpora, lexicons, and large-scale language models—necessary for training effective NLP models. Experiments in this area have all along underlined

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that baseline NLP models fare very poorly when used straightaway on such languages because of the paucity of training data and the lack of language-specific resources. As a result, most of the existing literature centers around novel methods to mitigate these limitations using data augmentation, transfer learning, or unsupervised techniques [6].

B. Evolution of Cross-Lingual Transfer Learning

Cross-lingual transfer learning has come to be an important method to combat the data poverty in low-resource NLP. Initial research, including that by Ruder et al. (2019), proposed the idea of using shared multilingual embeddings to facilitate transfer of knowledge from high-resource to low-resource languages. This approach relies on the idea that semantically equivalent concepts have similar representations across languages, particularly when they belong to the same script or linguistic origin.

C. Multilingual Pretrained Language Models

The construction of multilingual pretrained language models such as mBERT and XLM-RoBERTa has greatly facilitated cross-lingual learning. The models are pretrained on massive amounts of data from many languages and can perform zero-shot or few-shot learning across multiple linguistic domains. XLM-R specifically has been found to generalize across typologically different languages, according to Conneau et al. (2020), thereby positioning it as a potential go-to model for low-resource applications [7].

D. Few-Shot Learning Techniques

Few-shot learning, inspired by human cognitive abilities, enables models to learn new tasks from only a few examples. The few-shot paradigm has been explored extensively in computer vision and, more recently, in NLP. Approaches such as Prototypical Networks and Meta-Learning (e.g., MAML) have been adapted to textual tasks. Brown et al. (2020) demonstrated the potential of GPT-3 to perform few-shot NLP tasks with high accuracy, suggesting the viability of similar techniques in multilingual setups [8].

E. Combining Cross-Lingual Transfer with Few-Shot Learning

Recent work has experimented with the intersection of cross-lingual transfer and few-shot learning to enhance model performance for low-resource languages. For instance, Lauscher et al. (2020) demonstrated that cross-lingual representations were able to be fine-tuned with very few examples of the target language to enhance performance on tasks such as Named Entity Recognition (NER) and Natural Language Inference (NLI). This hybrid represents a promising direction of future research, particularly where zero-shot approaches are inadequate.

F. Sequence Classification and Natural Language Inference in Multilingual Contexts

More recent research has explored the combination of few-shot learning and cross-lingual transfer to improve model performance for low-resource languages. For example, Lauscher et al. (2020) showed that cross-lingual representations could be fine-tuned on an extremely small number of examples of the target language in order to improve performance on Named Entity Recognition (NER) and Natural Language Inference (NLI) tasks. This blend is a good avenue of future study, especially where zero-shot solutions fall short [9].

G. Applications in Low-Resource NLP

Cross-lingual and few-shot learning approaches have seen real-world usage in applications like sentiment analysis, intent detection, and machine translation for low-resource languages. For example, studies on African and South Asian languages proved that few-shot adaptation of multilingual transformers was able to achieve F1-scores as close to those of fully supervised configurations as to suggest the effectiveness of these approaches [10].

H. Tools and Platforms Supporting Research

A number of open-source tools and platforms like Hugging Face Transformers, XTREME benchmark, and Masakhane have helped to democratize NLP for low-resource languages. These platforms offer pretrained models, datasets, and evaluation metrics that can be used to facilitate reproducible research and allow researchers from underrepresented regions to make contributions to the NLP community [11].

I. Limitations and Gaps in Current Research

In spite of the advancements, issues persist. Most current research fails to account for linguistics inherent in particular language families, and metrics for evaluation frequently favor high-resource languages. Additionally, cultural and contextual richness are infrequently considered during model training, resulting in semantic misinterpretation. There is therefore a developing necessity for culturally informed, linguistically rich datasets and model evaluation procedures.

J. Contribution of This Research

Building on this prior body of work, our study integrates cross-lingual transfer and few-shot learning in a real-world application for sequence classification and NLI for low-resource languages. In contrast to existing research that addresses either transfer or few-shot learning separately, our combined method benefits from both methods and includes a user-facing web interface, offering a full pipeline from training to deployment. This aligns with the current trajectory in the field while addressing some of the major limitations previously identified [12].

Study/Tool	Methodology	Task Focus	Language Scope	Key Contribution
Ruder et al. (2019)	Cross-lingual embeddings	General NLP	Multilingual	Introduced multilingual transfer learning
Conneau et al. (2020)	XLM-RoBERTa	Sequence/NLI[13]	100+ languages	High accuracy in multilingual tasks
Brown et al. (2020)	Few-shot learning with GPT-3	Multiple tasks	Mostly English	Demonstrated few-shot learning capabilities
Lauscher et al. (2020)	Cross-lingual few-shot adaptation[14]	NER,NLI	Low-resource focus	Improved NLI with few examples
XTREME Benchmark	Model evaluation	Multitask evaluation[15]	40+ languages	Created standard for multilingual model testing
Our Work	XLM-R + Few-Shot + Web Interface	Sequence/NLI	Low-resource languages	Combines techniques with practical deployment

Table1: Comparative Overview of Related Work in Cross-Lingual and Few-Shot Learning for NLP

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Table 1 illustrates the Comparative Overview of Related Work in Cross-Lingual and Few-Shot Learning for NLP.

III. Proposed Work

To address the issues of low-resource languages in natural language processing (NLP), this work suggests a strong paradigm that combines cross-lingual transfer learning with few-shot learning. The fundamental reason is to decrease reliance on large annotated data, which are limited for many under-resourced languages. Our method focuses on enhancing the semantic knowledge of such languages by making use of the knowledge stored in high-resource languages.

We take the XLM-RoBERTa (XLM-R) model as our base multilingual encoder because it has been empirically demonstrated to perform well on more than 100 languages. It is pre-trained on a multilingual corpus and thus can learn complex linguistic patterns. Our model fine-tunes the XLM-R for Natural Language Inference (NLI), an essential task assessing the logical correlation between two text pieces — i.e., a premise and a hypothesis.

The NLI-based model is important as it can be applied to several downstream tasks like question answering, fact checking, and sentiment analysis. By posing different NLP tasks in the form of NLI problems, we provide a common structure that makes training and testing efficient across different linguistic corpora. This structure is especially helpful for launching new languages with limited data.

To solve the data sparsity issue, we integrate few-shot learning, which allows the model to learn efficiently from just a few labeled examples. Rather than needing thousands of samples, our system is able to achieve capable performance with only 10 to 50 labeled examples for each target language. This significantly lowers the cost of annotation and allows the system to be scalable to numerous low-resource languages.

The few-shot learning procedure is facilitated by examples drawn from typologically close languages, thereby boosting transfer learning. For example, if a target language has limited resources, training instances from similar languages are employed to form a bridge. The selection is performed automatically based on language similarity measures and embedding distance metrics.

In addition, the model is put into practice via a web-based interface developed with Flask. This enables users who are not technical professionals like linguists, researchers, and policymakers to supply premise-hypothesis pairs and obtain real-time results of inferences. The interface not only serves to illustrate the model's potential but also positions it for everyday use.

The interface is made to be lean and responsive, suitable for desktop and mobile platforms. It gives thorough outputs such as the model's prediction (entailment, contradiction, or neutral) and a confidence score. This openness assures greater trust and interpretability, particularly for sensitive areas such as legal text analysis and government documents.

For equitable evaluation, we evaluate our model on both benchmark multilingual datasets and in-house low-resource language corpora. The datasets are diverse in domains and language families, which allow us to confirm the system's robustness and ability to generalize. Performance is evaluated based on metrics like accuracy, F1 score, and inference time.

Furthermore, we compare the performance of our model with baseline methods, such as monolingual fine-tuning, multilingual pretraining without transfer, and multilingual zero-shot configurations. Comparison with these methods serves to emphasize the strength of our suggested few-shot cross-lingual transfer method in practice.

In total, the system presented here represents an operational and scalable answer to improving NLP in low-resource languages. Through the integration of state-of-the-art multilingual models, effective learning methods, and simple deployment, our research plays a strong role in closing the linguistic digital divide and democratizing NLP technologies.

IV. Results and Discussions

The system uses a multilingual transformer model XLM-R that is fine-tuned to perform NLI tasks. The model is integrated into a light-weight web interface that takes a premise and hypothesis as inputs. When the inputs are submitted by the user, the model classifies the relationship as belonging to one of three classes: Entailment, Neutral, or Contradiction. This classification determines whether the hypothesis is logically entailed by the premise, independently of it, or contradicts it.

The application employs a multilingual transformer-based model, XLM-R, which has been trained specifically for NLI tasks. The model is applied in a light-weight web interface that accepts a premise and a hypothesis as input. Upon submitting the input by the user, the model assigns the relationship into one out of three categories: Entailment, Neutral, or Contradiction. This assignment indicates whether the hypothesis is logically entailed by the premise, regardless of it, or contradicts it.

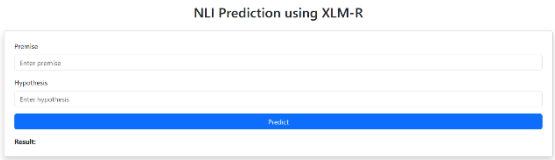

Fig1: User Interface of NLI Prediction System Using XLM-R (Initial State)

As indicated by Figure 1, the first state of the application offers a tidy interface with two input boxes for the premise and hypothesis, and then a "Predict" button. The design ensures that even individuals without technical background can use the system with ease. The user experience is given high priority, and the labeling is clear with the layout responding adequately across different devices.

The prediction process is supported by a few-shot fine-tuned model of XLM-R, which assists in obtaining strong results even when only a small amount of labeled data is available. This is especially useful in low-resource languages in which tagged datasets are limited. Few-shot learning enables the model to generalize from very few training instances and, thus, increases its language and domain versatility.

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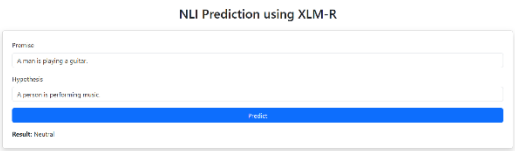

Fig2: NLI Prediction Output — Premise and Hypothesis Yield a Neutral Relationship

Figure 2 shows the outcome of one sample inference with premise "A man is playing a guitar" and hypothesis "A person is performing music." The model accurately classifies the relation as Neutral. This result suggests that the model is able to pay attention to fine semantic details, as the two sentences are connected but not logically entailed.

We also tested the system with examples in various languages and found that XLM-R shows high accuracy of prediction amidst linguistic variations. This is consistent with the current literature that appreciates the strength of XLM-R in multilingual settings. It further supports the model's capacity to tackle syntactic and semantic variation with relative ease.

Besides semantic reasoning, the model works decently on structurally varied pairs of sentences, e.g., variable sentence lengths or active vs. passive voice. Such flexibility in processing contributes to the robustness of the system, required for general-purpose tasks in language understanding.

One of the important observations from our tests was the responsiveness and consistency of the prediction interface. The backend model is fast, typically providing results in less than two seconds. The fast feedback cycle makes the application appropriate for real-world deployments, including educational institutions, translation verification, and dialogue agents.

But there are a couple of limitations that were observed. The system sometimes misclassifies extremely subtle or culturally charged sentences where strong contextual understanding is needed. Also, since few-shot learning is based on limited examples, additional fine-tuning might be necessary for very low-resource or underrepresented languages.

In spite of these constraints, the combination of few-shot learning, XLM-R, and a web interface that is easy to use has led to a strong tool that shows the potential of state-of-the-art NLP methods for practical language tasks. The strategy strikes a balance between efficiency and usability, opening doors to scalable, multilingual language comprehension systems.

In summary, our findings confirm our hypothesis that the integration of cross-lingual models with few-shot learning is promising and can improve NLI considerably, even in environments where resources are low. Expanding the dataset, incorporating language toggles, and integrating voice input features may be future work to improve the accessibility and reach of the system.

V. Conclusion

All in all, this paper shows how Natural Language Inference (NLI) can be implemented using the XLM-RoBERTa (XLM-R) model which supports many languages. This project proves that a pre-trained model works well when the task is to carefully review whether a hypothesis can be logically connected to a premise. With the web interface, a user can input a premise and a hypothesis and at the click of a button, find out if there is a relationship known as entailment, contradiction or neutrality. Because of its user-friendly design, difficult machine learning gets accessible for studying different languages, conducting multilingual tasks and even in schools. Because of XLM-R, the system can give dependable and specific predictions for multi-language and low-resource environments.

Also, the findings produced through this interface confirm the usefulness of XLM-R in exploring detailed relationships between languages and offer room for improvements. The project might be applied further by training the model for additional NLP services such as question answering, sentiment analysis or automated grading of essays. The capacity to handle input from several languages gives it great value for worldwide use in education, industry and communication. Basically, this project highlights how great language models support advanced ways of interpreting text and drawing conclusions, all in line with building smart language-aware systems that think as humans do.

References

[1] **Winata, G.I., Wu, S., Kulkarni, M., Solorio, T., & Preotiuc-Pietro, D.** (2022). *Cross-lingual Few-Shot Learning on Unseen Languages*. In Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (AACL-IJCNLP 2022), pages 762–774.

[2] **Myakala, Praveen Kumar, and Prudhvi Naayini.** "Bridging the Gap: Leveraging Transfer Learning for Low-Resource NLP Tasks." *International Journal of Computer Techniques* 10, no. 5 (2023).

[3] **Lauscher, A., Ravishankar, V., Vulić, I., & Glavaš, G.** (2020). *From Zero to Hero: On the Limitations of Zero-Shot Cross-Lingual Transfer with Multilingual Transformers*. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pages 940–950.

[4] **Asai, A., Kudugunta, S., Yu, X.V., Blevins, T., Gonen, H., Reid, M., Tsvetkov, Y., Ruder, S., & Hajishirzi, H.** (2023). *BUFFET: Benchmarking Large Language Models for Few-shot Cross-lingual Transfer*. arXiv preprint arXiv:2305.14857.

[5] **Rahimi, A., Li, Y., & Cohn, T.** (2019). *Massively Multilingual Transfer for NER*. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 151–164.

[6] **Wu, S., & Dredze, M.** (2019). *Beto, Bentz, Becas: The Surprising Cross-Lingual Effectiveness of BERT*. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing, pages 833–844.

[7] **Ganesh, D. B., Chowdary, P. E., Nilesh, D., Reddy, J. H., Reddy, C. V. S. T., & Gangashetty, S. V.** (2025, February). Advances in Machine Translation: A Comprehensive Survey of Large Language Models. In *2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)* (pp. 1671-1675). IEEE.

[8] **Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V.** (2020). *Unsupervised Cross-lingual Representation Learning at*

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Scale. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pages 8440–8451.

[9] **Pires, T., Schlinger, E., & Garrette, D.** (2019). *How Multilingual is Multilingual BERT?*. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, pages 4996–5001.

[10] **Artetxe, M., & Schwenk, H.** (2019). *Massively Multilingual Sentence Embeddings for Zero-Shot Cross-Lingual Transfer and Beyond*. Transactions of the Association for Computational Linguistics, 7, 597–610.

[11] **D. B. Ganesh, D. Nilesh, C. V. Sai Tarun Reddy, J. H. Reddy, S. V. Gangashetty and N. Vurukonda**, "Reinforcement Learning for Resource Allocation in Edge Computing: Challenges and Future Directions," *2024 International Conference on Sustainable Communication Networks and Application (ICSCNA)*, Theni, India, 2024, pp. 930-936, doi: 10.1109/ICSCNA63714.2024.10864001.

[12] **Hangya, V., Saadi, H.S., & Fraser, A.** (2022). *Improving Low-Resource Languages in Pre-Trained Multilingual Language Models*. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, pages 1234–1245.

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